# CARTESIAN SHEAVES AS DESCENT DATA ON A SIMPLICIAL SITE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: in-source Situation situation-simplicial-site; external stubs simplicial-remark-relations (Simplicial chapter), descent-section-descent-modules (Descent chapter).

*Editorial note.* The label lemma-characterize-cartesian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Simplicial face-map relations. Simplicial, Remark simplicial-remark-relations. the identities d^1_1 d^2_2 = d^1_1 d^2_1, d^1_1 d^2_0 = d^1_0 d^2_2, d^1_0 d^2_0 = d^1_0 d^2_1 that make the cocycle condition on alpha well defined.
- Descent datum construction pattern. Descent, Section descent-section-descent-modules. the tau^n_i and tau^n_{ij} degeneracy maps and the quasi-inverse construction pattern borrowed for sheaves on the simplicial site.

*Source reference.* Original section 023F: sources/stacks-chow/tags.txt, source lines 2740–2740. Prerequisite pointer invoked at proof line(s) 2613; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09WF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 953–953. Prerequisite pointer invoked at proof line(s) 2637; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original remark 016C: sources/stacks-chow/tags.txt, source lines 1539–1539. Prerequisite pointer invoked at proof line(s) 2589; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2562; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 09WE. The category of cartesian sheaves of sets (resp. abelian groups) is equivalent to the category of pairs $(\mathcal{F}, \alpha)$ where $\mathcal{F}$ is a sheaf of sets (resp. abelian groups) on $\mathcal{C}_0$ and*

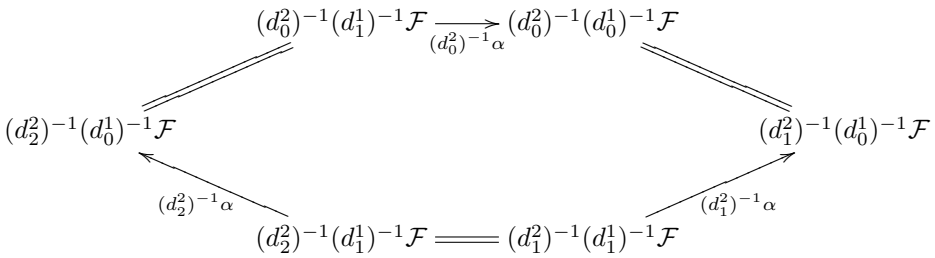
$$\alpha : (f_{\delta_1^1})^{-1}\mathcal{F} \longrightarrow (f_{\delta_0^1})^{-1}\mathcal{F}$$

*is an isomorphism of sheaves of sets (resp. abelian groups) on $\mathcal{C}_1$ such that $(f_{\delta_1^2})^{-1}\alpha = (f_{\delta_0^2})^{-1}\alpha \circ (f_{\delta_2^2})^{-1}\alpha$ as maps of sheaves on $\mathcal{C}_2$.*

*Proof.* We abbreviate $d_j^n = f_{\delta_j^n} : Sh(\mathcal{C}_n) \to Sh(\mathcal{C}_{n-1})$. The condition on $\alpha$ in the statement of the lemma makes sense because

$$d_1^1 \circ d_2^2 = d_1^1 \circ d_1^2, \quad d_1^1 \circ d_0^2 = d_0^1 \circ d_2^2, \quad d_0^1 \circ d_0^2 = d_0^1 \circ d_1^2$$

as morphisms of topoi $Sh(\mathcal{C}_2) \to Sh(\mathcal{C}_0)$, see Simplicial, Remark 016C. Hence we can picture these maps as follows




and the condition signifies the diagram is commutative. It is clear that given a cartesian sheaf $\mathcal{G}$ of sets (resp. abelian groups) on $\mathcal{C}_{total}$ we can set $\mathcal{F} = \mathcal{G}_0$ and $\alpha$ equal to the composition

$$(d_1^1)^{-1}\mathcal{G}_0 \to \mathcal{G}_1 \leftarrow (d_1^0)^{-1}\mathcal{G}_0$$

where the arrows are invertible as $\mathcal{G}$ is cartesian. To prove this functor is an equivalence we construct a quasi-inverse. The construction of the quasi-inverse is analogous to the construction discussed in Descent, Section 023F from which we borrow the notation $\tau_i^n : [0] \to [n]$, $0 \mapsto i$ and $\tau_{ij}^n : [1] \to [n]$, $0 \mapsto i$, $1 \mapsto j$. Namely, given a pair $(\mathcal{F}, \alpha)$ as in the lemma we set $\mathcal{G}_n = (f_{\tau_n^n})^{-1}\mathcal{F}$. Given $\varphi : [n] \to [m]$ we define $\mathcal{G}(\varphi) : (f_\varphi)^{-1}\mathcal{G}_n \to \mathcal{G}_m$ using

$$\begin{array}{ccccccc}
(f_\varphi)^{-1}\mathcal{G}_n & = & (f_\varphi)^{-1}(f_{\tau_n^n})^{-1}\mathcal{F} & = & (f_{\tau^m_{\varphi(n)}})^{-1}\mathcal{F} & = & (f_{\tau^m_{\varphi(n)m}})^{-1}(d_1^1)^{-1}\mathcal{F} \\
 & & & & & & \Big\downarrow{\scriptstyle (f_{\tau^m_{\varphi(n)m}})^{-1}\alpha} \\
\mathcal{G}_m & = & (f_{\tau_m^m})^{-1}\mathcal{F} & = & (f_{\tau^m_{\varphi(n)m}})^{-1}\mathcal{F} & = & (f_{\tau^m_{\varphi(n)m}})^{-1}(d_0^1)^{-1}\mathcal{F}
\end{array}$$

We omit the verification that the commutativity of the displayed diagram above implies the maps compose correctly and hence give rise to a sheaf on $\mathcal{C}_{total}$, see Lemma 09WF. We also omit the verification that the two functors are quasi-inverse to each other. □